Chapter 14 — Atmosphere

Complete Notes · PECTAA Chemistry 11 · Board-style longs + conceptual shorts

Topic 1 — Layers of atmosphere and air pollutants

Q.1 Describe the layers of the atmosphere and classify air pollutants.

The atmosphere has four main layers. The troposphere is the lowest layer, where weather occurs and temperature decreases with altitude. The stratosphere contains the ozone layer and temperature rises with altitude because ozone absorbs UV. The mesosphere is the coldest layer, and the thermosphere is the hottest because it absorbs high-energy radiation.

Air pollutants may be primary, emitted directly, such as CO, SO_2 and NO; or secondary, formed in the atmosphere, such as ozone and PAN. Major pollutants include CO, CO_2, NOx, SOx, hydrocarbons, CFCs, PAHs, POPs, VOCs, particulate matter and heavy metals.

Important Conceptual Short Questions — Topic 1

1. Why is the ozone layer located in the stratosphere important?

It absorbs harmful ultraviolet radiation from the Sun, protecting living organisms from DNA damage, skin cancer and ecosystem harm.

2. How do primary and secondary pollutants differ?

Primary pollutants are emitted directly from sources, while secondary pollutants form by chemical reactions in the atmosphere.

3. Why is the mesosphere the coldest layer?

It has very little ozone or other gases to absorb solar radiation, and temperature decreases with altitude up to the mesopause.

4. How do particulate matter and heavy metals affect air quality?

Particulates reduce visibility and penetrate the respiratory system, while heavy metals such as Pb, Hg and Cd are toxic and can accumulate in organisms.

Topic 2 — Sources and major pollutants

Q.2 Explain natural and human sources of air pollution and the chemistry of major pollutants.

Natural sources include volcanoes, dust storms, forest fires, lightning, pollen and microbial activity. Human sources include vehicles, industries, power plants, agriculture, waste burning and deforestation. Mobile sources move, while point sources are fixed.

CO forms by incomplete combustion and binds haemoglobin strongly. CO_2 is a greenhouse gas. NOx forms in high-temperature combustion and contributes to smog and acid rain. SO_2 comes from sulfur-containing fuels and forms acid rain.

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CFCs release chlorine radicals in the stratosphere, which catalytically destroy ozone.

Important Conceptual Short Questions — Topic 2

1. Why does incomplete combustion produce CO?

When oxygen is limited, carbon is oxidized only to CO instead of CO_2.

2. How do CFCs destroy stratospheric ozone?

UV light breaks CFCs to release chlorine radicals. A chlorine radical reacts with O_3 and is regenerated, allowing one radical to destroy many ozone molecules.

3. Why is ground-level ozone harmful while stratospheric ozone is protective?

Stratospheric ozone absorbs harmful UV radiation. Ground-level ozone is a strong oxidant that irritates lungs, damages plants and attacks materials.

4. How do vehicles contribute to NOx pollution?

High temperatures in engines make nitrogen and oxygen from air react to form NO, which can oxidize further to NO_2.

Topic 3 — Human impacts, smog, acid rain and greenhouse effect

Q.3 Explain the impact of human activities on the atmosphere, including smog, acid rain and global warming.

Burning fossil fuels releases CO_2, CO, NOx, SO_2 and particulates. Deforestation reduces CO_2 uptake and releases stored carbon when trees are burned or decay. Photochemical smog forms when NOx and VOCs react in sunlight, producing ozone, aldehydes and PAN.

Acid rain forms when SO_2 and NOx are oxidized and dissolved in rainwater to form H_2SO_4 and HNO_3. It damages buildings, soils, lakes and vegetation. The greenhouse effect occurs when gases such as CO_2, CH_4 and N_2O absorb infrared radiation emitted by Earth. Enhanced greenhouse effect causes global warming.

Important Conceptual Short Questions — Topic 3

1. Why does acid rain damage marble buildings?

Marble contains $CaCO_3$, which reacts with acids to form soluble salts, CO_2 and water, so the surface is eroded.

2. How does deforestation increase atmospheric CO_2?

Fewer trees remove CO_2 by photosynthesis, and burning or decay of cleared vegetation releases stored carbon as CO_2.

3. Why is photochemical smog common in sunny cities with heavy traffic?

Traffic emits NOx and VOCs, and strong sunlight provides the energy needed for the reactions that form ozone and PAN.

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4. How do greenhouse gases warm the Earth?

They allow much solar radiation to enter but absorb and re-emit infrared radiation from Earth's surface, trapping heat in the lower atmosphere.

Topic 4 — Air quality, health, monitoring and control

Q.4 Discuss air quality, health effects, monitoring methods and strategies to reduce air pollution.

Air quality is often expressed by the Air Quality Index, AQI, which summarizes concentrations of key pollutants. Poor air quality causes respiratory and cardiovascular disease, cancer risk and developmental effects. Fine particulates, $PM_{2.5}$, are especially dangerous because they penetrate deep into the lungs and can enter the bloodstream.

Air quality is monitored by continuous emission monitoring systems, fixed monitoring stations, remote sensing and instruments such as nephelometers for particulates. Control strategies include cleaner fuels, catalytic converters, scrubbers, selective catalytic reduction, public transport, renewable energy and laws such as PEPA 1997 and NEQS.

Textbook check (p. 308): a nephelometer measures light scattering by particulate matter; AQI is a calculated index based on several pollutant measurements, not a single direct nephelometer reading.

Important Conceptual Short Questions — Topic 4

1. Why are fine particulates more dangerous than coarse particles?

$PM_{2.5}$ can reach the alveoli and may enter the bloodstream, causing inflammation and cardiovascular effects, while larger particles are often trapped in the upper airways.

2. How does CO reduce the oxygen-carrying capacity of blood?

CO binds haemoglobin much more strongly than oxygen, forming carboxyhaemoglobin and preventing oxygen transport.

3. Why is air-quality monitoring important for public health?

It identifies pollution levels and trends, warns vulnerable groups, and guides control measures to reduce exposure.

4. How can renewable energy reduce atmospheric pollution?

Solar, wind and hydro power generate electricity without burning fossil fuels, reducing emissions of CO_2, SO_2, NOx and particulates.

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